POSLink Integration Setup Guide

Version: 1.04

PAX Technology, Inc.

Preface

POSLink Integration Setup Guide
Version: V1.02
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REVISION HISTORY

Date	Version	Author	Description
2018/03/14	V1.00	KyleL	Initial Release
2018/06/08	V1.01	KyleL	Added A Series and E Series
2018/06/11	V1.02	NickZ	Added iOS POSLink
2018/08/13	V1.03	KyleL	Updated HTTPS/SSL Requirements
2019/03/14	V1.04	KyleL	Added BT for Android POSLink

1. General Information

1.1 Document Purpose

This document provides the detailed description for all user functions, which can be performed on the POS terminal. The document is intended for use by the Software Engineers developing and maintaining the Payment Application design for the US Payment Card Industry, also for use by customer and Independent Sales Vendor (ISV). The document should be maintained along with the application itself.

It is required that this document is stored in a safe place for future reference and modifications.

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2. SDK Information

2.1 POSLink

- Four SDK versions:
  - .NET
  - iOS
  - Java
  - Android
- Support both Semi-Integration and Full-Integration APIs
- Support USB, Serial Port, TCP, SSL/TLS, HTTP, HTTPs and Bluetooth ECR communication types.
- Only POSLink Android SDK supports AIDL (Android Internal Communication Type) ECR communication type. AIDL is for A920, A60, etc. devices (Android terminals).
- The E-series (E500, E800) uses Android version with USB only

2.2 POSDK

- Support Android only
- Support semi-integration only
- Support AIDL ECR communication type only
- The UI of the payment application is handled by the third party's POS application.

3. ECR/POS Setup

3.1 PC .NET POSLink

Four versions of PC .NET POSLink are normally provided; Install according to the system architecture, project settings and target framework:

a. .NET3.5 x86
b. .NET3.5 x64
c. .NET4.0 x86
d. .NET4.0 x64

1. Unzip POSLink_V1.xx.xx_20xxxxxx_InstallationPackage(x86) for x86 based system architecture or POSLink_V1.xx.xx_20xxxxxx_InstallationPackage(x64) for x64 based system architecture.

POSLink_V1.01.04_20171120_InstallPackage(x86)	10/20/2017 4:22 PM	File folder	
POSLink For PC_V1.01.02_20161014.zip	5/24/2017 2:14 PM	WinRAR ZIP archive	27,261 KB
POSLink For PC_V1.01.03_20170406.zip	5/24/2017 2:14 PM	WinRAR ZIP archive	27,868 KB
POSLink For PC_V1.01.04_20171120.zip	12/12/2017 10:52 ...	WinRAR ZIP archive	27,384 KB
POSLink_V1.01.04_20171120_InstallPackage(x64).rar	11/22/2017 10:11 ...	WinRAR archive	7,803 KB
POSLink_V1.01.04_20171120_InstallPackage(x86).rar	11/22/2017 10:11 ...	WinRAR archive	6,543 KB
POSLink_V1.01.04_20171120_Samples.rar	11/22/2017 10:11 ...	WinRAR archive	9,305 KB
POSLinkII_V1.01.04_20171201 API Guide.pdf	12/12/2017 11:53 ...	PDF File	4,556 KB
Revision History.txt	12/12/2017 10:54 ...	Text Document	11 KB

2. Inside the unzipped folder, choose the folder based on project's target framework.

.net3.5	11/20/2017 5:57 PM	File folder
.net4.0	11/20/2017 3:37 PM	File folder

3. Execute POSLinkSetup.exe and proceed through the installation process

4. By default, POSLink.dll and dependencies can be located at:
For x86: C:\Program Files\PAX\POSLink
For x64: C:\Program Files(x86)\PAX\POSLink
5. Import/Reference POSLink.DLL into project.

6. Instantiate POSLink, CommSetting, ManageRequest, ManageResponse and ProcessTransResult.
7. Assign appropriate commSetting variables.
8. Assign CommSetting variables to POSLink Object.

3.1.1 Sample Code

```
public static void Main()
        {
        private POSLink.PosLink poslink;
        private POSLink.CommSetting commSetting;
        private POSLink.ManageRequest manageReq;
        private POSLink.ManageResponse manageRes;
        private POSLink.ProcessTransResult transResult;

        /*Instantiating Objects */
        poslink = new POSLink.PosLink();
        commSetting = new POSLink.CommSetting();
        manageReq = new POSLink.ManageRequest();

        /*
        * Setting up Communications
        */
        commSetting.CommType = "TCP";
        commSetting.TimeOut = "30000";
        commSetting.DestIP = "172.20.2.115";
        commSetting.DestPort = "10009";
        poslink.CommSetting = commSetting;

        /*Sets up ManageRequest Object to perform INIT Command*/
        manageReq.TransType = manageReq.ParseTransType("INIT");

        /*Assign ManageRequest Object to the POSLink Object */
        poslink.ManageRequest = manageReq;

        /*Tell terminal to perform the assigned transaction */
        transResult = poslink.ProcessTrans();

        /*Stores the response from the terminal */
        manageRes = poslink.ManageResponse;

        /*Prints stored information from terminal */
        Console.WriteLine(manageRes.ResultCode + " - " + manageRes.ResultTxt
        + "\r\nSN: " + manageRes.SN
        + "\r\nModel Name: " + manageRes.ModelName
```

```
        + "\r\nOS Version: " + manageRes.OSVersion
        + "\r\nMac Address: " + manageRes.MacAddress
        + "\r\nLines Per Screen: " + manageRes.LinesPerScreen
        + "\r\nCharsPerLine: " + manageRes.CharsPerLine);
        }
```

3.2 JAVA POSLink

1. Import/Reference POSLink_xxxxxx.jar into project.
2. Instantiate POSLink, CommSetting, ManageRequest, ManageResponse and ProcessTransResult.
3. Assign appropriate commSetting variables.
4. Assign CommSetting variables to POSLink Object.

3.2.1 Sample Code

```
public static void main(String args[]) throws IOException
    {

        CommSetting commSetting = new CommSetting();
        commSetting .setType(CommSetting.TCP);
        commSetting .setDestIP("192.168.000.101");
        commSetting .setDestPort("10009");
        commSetting .setTimeOut("20000");


        PosLink poslink = new PosLink();
        poslink.SetCommSetting(commSetting);

        //////////////////MANAGE COMMAND START///////////////////////////////

         ManageRequest manageReq = new ManageRequest();
         manageReq.TransType = manageReq.ParseTransType("INIT");
         poslink.ManageRequest = manageReq;
         ProcessTransResult transResult = poslink.ProcessTrans();

         System.out.println(transResult.Code + ":" + transResult.Msg);

         ManageResponse manageRes = poslink.ManageResponse;
         if (r != null)
         System.out.println(manageRes.ResultCode + " - " + manageRes.ResultTxt
        + "\r\nSN: " + manageRes.SN
        + "\r\nModel Name: " + manageRes.ModelName
        + "\r\nOS Version: " + manageRes.OSVersion
        + "\r\nMac Address: " + manageRes.MacAddress
```

```
        + "\r\nLines Per Screen: " + manageRes.LinesPerScreen
        + "\r\nCharsPerLine: " + manageRes.CharsPerLine);
    }
```

3.3 iOS POSLink

1. Import/Reference libPOSLink.a into project.
2. Instantiate POSLink, CommSetting, ManageRequest, ManageResponse and ProcessTransResult.
3. Assign appropriate commSetting variables.
4. Assign CommSetting variables to POSLink Object.

3.3.1 Sample Code

```
- (void)main
{
      CommSetting *comm = [[CommSetting alloc] init];
      POSLink *poslink = [[POSLink alloc] initWithCommSetting:comm];

      /*Setting up Communications */
      poslink.commSetting.commType = @"TCP"
      poslink.commSetting.timeout = @"20000"
      poslink.commSetting.destIP = @"192.168.98.101"
      poslink.commSetting.destPort = @"10009"

      /* MANAGE COMMAND*/
      ManageRequest *manageReq = [[ManageRequest alloc] init];
      manageReq.TransType = [ManageRequest ParseTransType:@"INIT"];
      poslink.manageRequest = manageReq;

      ProcessTransResult *ret = [poslink processTrans:MANAGE];
      NSLog(@"%@,%@",ret.code,ret.msg);

      ManageResponse *manageRes = [[ManageResponse *alloc] init];
      manageRes = poslink.manageResponse;

      NSLog(@"ResusltCode:%@,ResultTxt:%@,SN:%@,ModelName:%@,OSVersion:%@,
      MacAddress:%@,LinesPerScreen:%@,CharsPerLine:%@",manageRes.ResultCode,
      manageRes.ResultTxt,manageRes.SN,manageRes.ModelName,manageRes.OSVersion,
      manageRes.MacAddress, manageRes.LinesPerScreen,manageRes.CharsPerLine);
}
```

4. CommSetting Setup

4.1 UART

Note: For Windows and Linux Only. This requires USB Drivers to be installed. The USB Drivers will simulate USB Port as a Serial Port. POSLink will treat the USB connection as a serial connection. UART communication type should be when using serial cable or USB cable.

```
//Configure Communication Settings
            //Note: Settings will vary depending on your usage
            commSetting.CommType = "UART";
            commSetting.SerialPort = "COM4";//Check System Configuration to
Confirm the Serial Port the terminal is connected to
            commSetting.BaudRate = "115200";//
            commSetting.TimeOut = "30000";//in milliseconds; connection will be
timed out after 30 seconds of failed connection
```

4.2 TCP

```
            //Configure Communication Settings
            //Note: Settings will vary depending on your usage
            commSetting.CommType = "TCP";
            commSetting.DestIP = "192.168.5.123";
            commSetting.DestPort = "10009";
            commSetting.TimeOut = "30000"; //in milliseconds; connection will be
timed out after 30 seconds of failed connection
```

4.3 HTTP/HTTPS

Note: HTTPs requires SSL Certificates. See Section 6 SSL Communication Setup.

```
            //Configure Communication Settings
            //Note: Settings will vary depending on your usage
            commSetting.CommType = "HTTP"; //or HTTPS
            commSetting.DestIP = "192.168.5.164";
            commSetting.DestPort = "10009";
            commSetting.TimeOut = "30000";//in milliseconds; connection will be
timed out after 30 seconds of failed connection
```

4.4 SSL

Note: Requires SSL Certificates. See Section 6 SSL Communication Setup.

```
            //Configure Communication Settings
            //Note: Settings will vary depending on your usage
```

```
            commSetting.CommType = "SSL";
            commSetting.DestIP = "192.168.5.164";
            commSetting.DestPort = "10009";
            commSetting.TimeOut = "30000";//in milliseconds; connection will be
timed out after 30 seconds of failed connection
```

4.5 BLUETOOTH

Note: Bluetooth is treated as a Serial Port by POSLink on PC.

```
            //Configure Communication Settings
            //Note: Settings will vary depending on your usage
            commSetting.CommType = "UART";
            commSetting.SerialPort = "COM24";//Check System Configuration to
Confirm the Serial Port the terminal is connected to
            commSetting.TimeOut = "30000";//in milliseconds; connection will be
timed out after 30 seconds of failed connection
```

On Android devices, using Java POSLink, Bluetooth can be selected as CommType.

```
            //Configure Communication Settings
            //Note: Settings will vary depending on your usage
            commSetting.setType(CommSetting.BT);
            commSetting.setMacAddr("12:34:56:78");
            commSetting.setTimeOut("30000"); //in milliseconds;
            //connection will be timed out after 30 seconds of failed
            //connection
```

4.6 USB – Android Devices/PAX E Series

```
CommSetting commSetting = new CommSetting();
commSetting.setType(CommSetting.USB);
commSetting.setTimeOut("30000");
commSetting.setEnableProxy(true);   //Enables Communication with Q20 terminal
```

4.7 AIDL – PAX A Series

```
CommSetting commSetting = new CommSetting();
commSetting.setType(CommSetting.AIDL);
commSetting.setTimeOut("3000");
```

5. PAX Terminal Setup

Note: All terminals must have a PAX payment application installed. An application can be installed from BroadPOS TMS or PAXStore.

5.1 A920

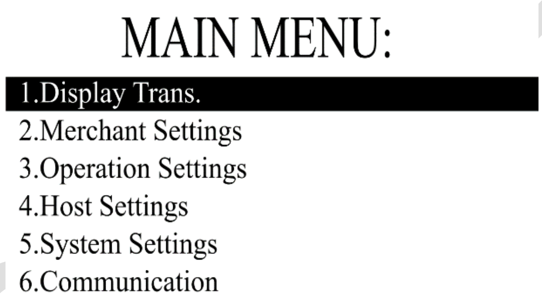
5.1.1 AIDL

A920 currently only supports AIDL communication type. See Section 4.7 AIDL -PAX A Series.

5.2 D200

5.2.1 Access Communication Menu

1. Press Enter Key to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Scroll down using F2 Key.
4. Select Communication.



5. Enter same password as above.

5.2.2 UART/COM

1. Access Communication Menu.
2. Scroll down using F2 Key.
3. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

4. Select COM1.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.Bluetooth

5. Select Baud Rate (9600 or 115200).

ECR BaudRate:1

1.9600
2.115200

5.2.3 Ethernet - TCP/IP, HTTP, HTTPS, SSL

Note: D200 communicates over Ethernet via WIFI Protocol.

1. Access Communication Menu
2. Scroll down using the F2 Key F2.
3. Select WiFi Parameters

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

4. Select SSID. Or if previously configured, scroll down using the F2 Key F2 and select Switch Router

WIFI PARAMS

1.SSID
2.Security
3.Password
4.DHCP Type
5.IP Address
6.Subnet Mask

5. Choose the SSID of the WIFI router.
6. Select Password and enter Access Point Preshared Key
7. Press Cancel Key
8. Scroll down twice using the F2 Key F2.
9. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

10. Select Ethernet.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.Bluetooth

11. Enter Communication Port Number. Default port is 10009.

COMM. PORT:
10009

12. Choose Communication Protocol (TCP/IP, HTTP, HTTPS, SSL)

PROTOCOL: 1

1.TCP/IP
2.HTTP GET
3.SSL
4.HTTPS GET

5.2.4 USB

1. Access Communication Menu
2. Scroll down twice using the F2 Key F2.
3. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

4. Select USB.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.Bluetooth

5.2.5 Bluetooth

1. Access Communication Menu
2. Scroll down twice using the F2 Key F2.
3. Select BlueTooth Params.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

4. Take note of the Device Name and PinCode.

BLUETOOTH PARAM

1.Device Name
2.PinCode

5. Press Cancel Key
6. Scroll down twice using the F2 Key F2.
7. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Wireless Parameters
7.WiFi Parameters
8.BlueTooth Params
9.ECR Comm. Type

8. Select Bluetooth.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.Bluetooth

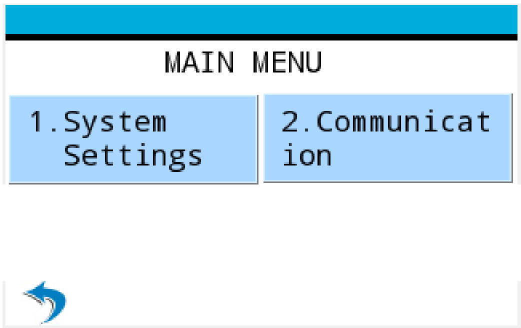
9. Device is Now Discoverable by other devices. Look for Device Name as noted above and enter PIN code if required.

5.3 D210s

5.3.1 Access Communication Menu

1. Press Menu Key to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Scroll down using the Down Arrow Key .

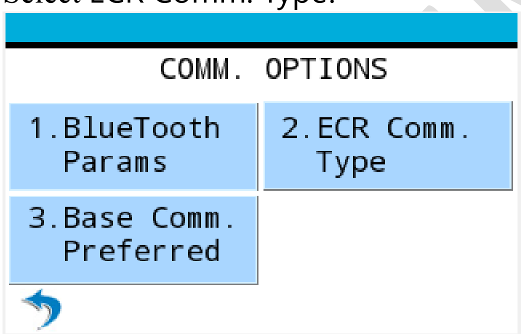
4. Select Communication.



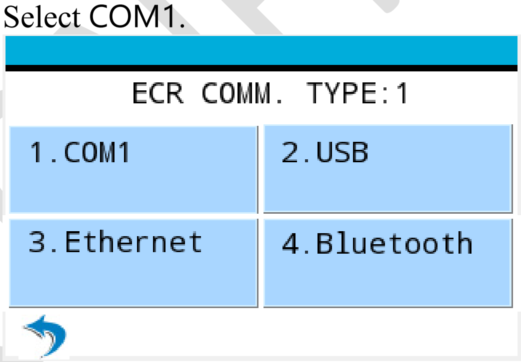
5. Enter same password as above.

5.3.2 UART/COM

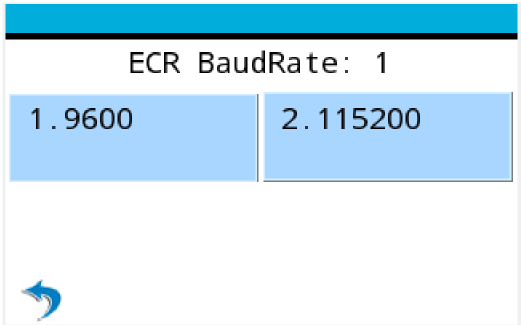
1. Access Communication Menu
2. Scroll down twice using the Down Arrow Key.
3. Select ECR Comm. Type.



4. Select COM1.



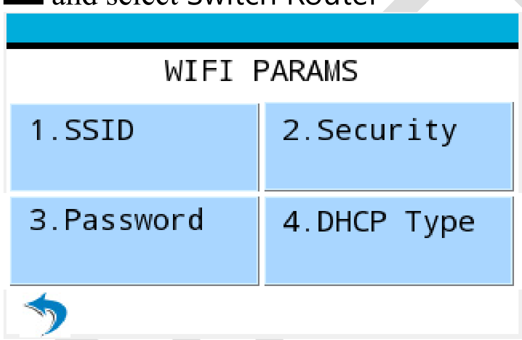
5. Select Baud Rate (9600 or 115200).



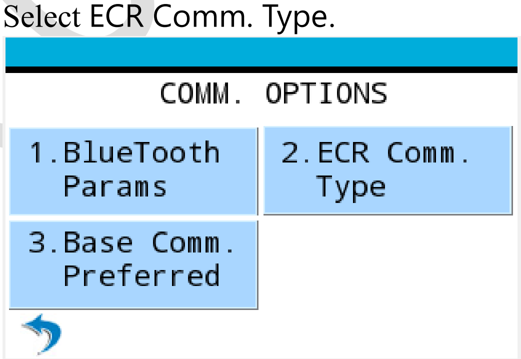
5.3.3 Ethernet – TCP/IP, HTTP, HTTPS, SSL

Note: D210 communicates over Ethernet via WIFI Protocol.

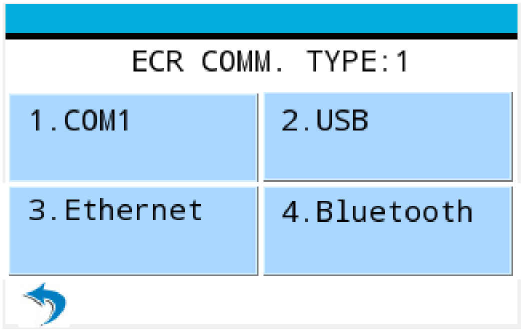
1. Access Communication Menu
2. Scroll down using the Down Arrow Key .
3. Select WiFi Parameters
4. Select SSID. Or if previously configured, scroll down using the Down Arrow Key and select Switch Router



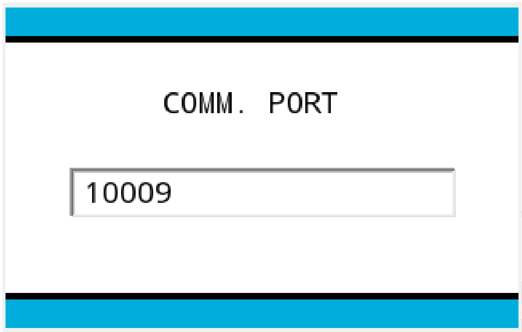
5. Choose the SSID of the WIFI router.
6. Select Password and enter Access Point Preshared Key
7. Press Cancel Key
8. Scroll down twice using the Down Arrow Key .
9. Select ECR Comm. Type.



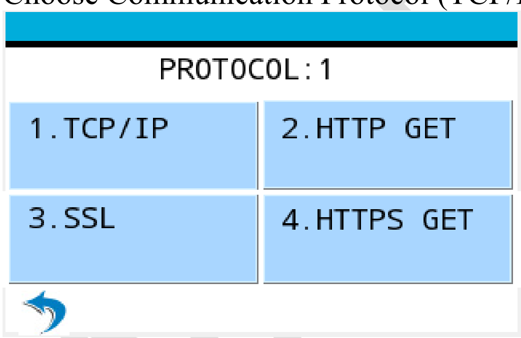
10. Select Ethernet.



11. Enter Communication Port Number. Default port is 10009.

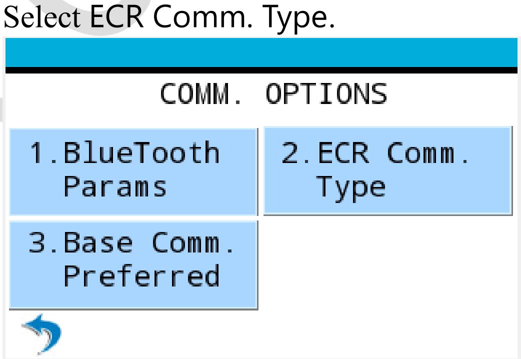


12. Choose Communication Protocol (TCP/IP, HTTP, HTTPS, SSL)

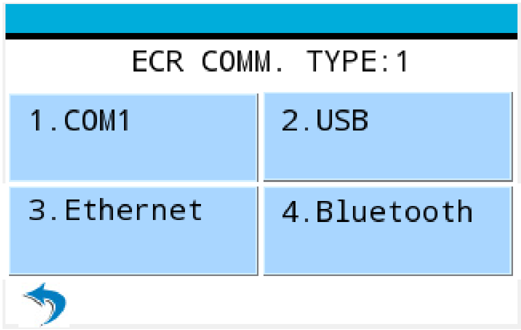


5.3.4 USB

1. Access Communication Menu
2. Scroll down twice using the Down Arrow Key .
3. Select ECR Comm. Type.

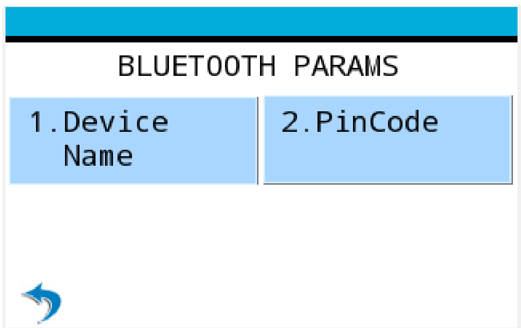


4. Select USB.

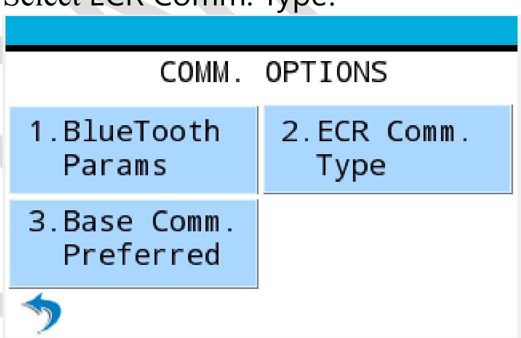


5.3.5 Bluetooth

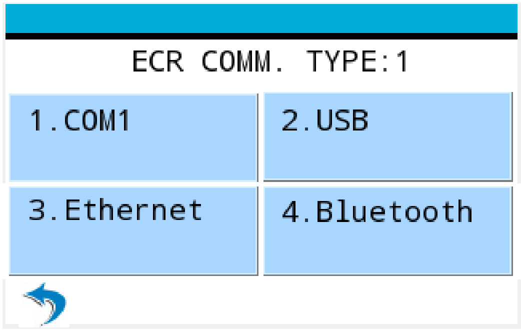
1. Access Communication Menu
2. Scroll down twice using the Down Arrow Key.
3. Select BlueTooth Params.



4. Take note of the Device Name and PinCode.
5. Press Cancel Key
6. Scroll down twice using the Down Arrow Key.
7. Select ECR Comm. Type.



8. Select Bluetooth.



9. Device is Now Discoverable by other devices. Look for Device Name as noted above and enter PIN code if required.

5.4 E500

5.4.1 USB

Add Following to AndroidManifest.xml

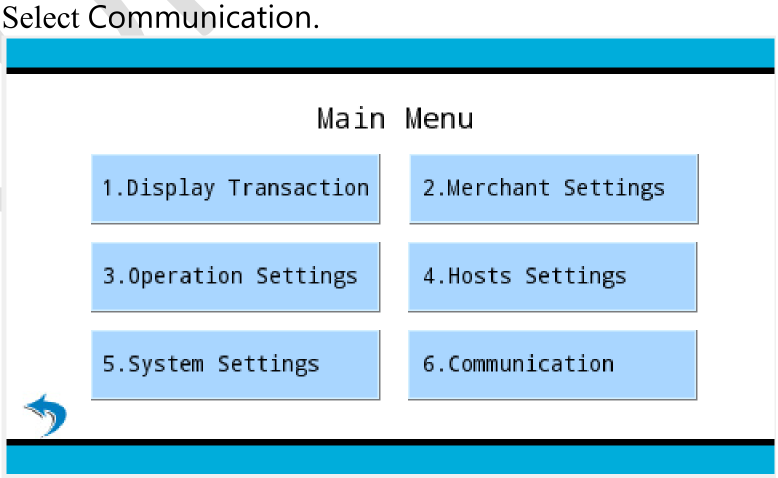
```
<uses-feature android:name="android.hardware.usb.host"/>
```

E500 currently only supports USB communication type. See Section 4.7 USB – Android Device/PAX E Series.

5.5 MT30/MT30s

5.5.1 Access Communication Menu

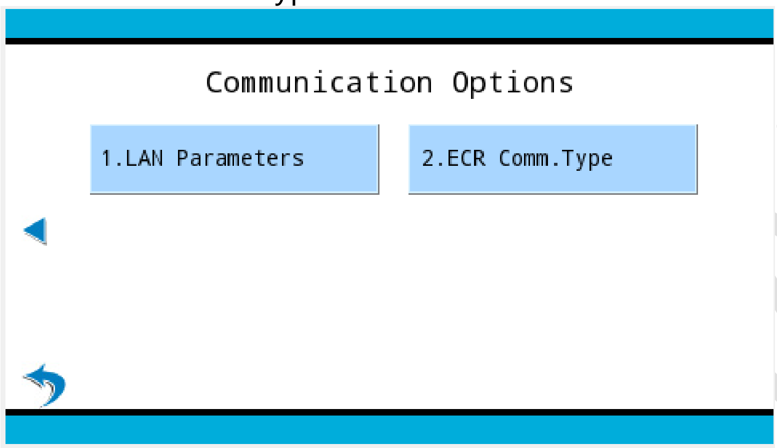
1. Press Enter Key to access Main Menu.
2. Enter Password. Default Password is 1 or today’s date (format: mmddyyyy).
3. Select Communication.



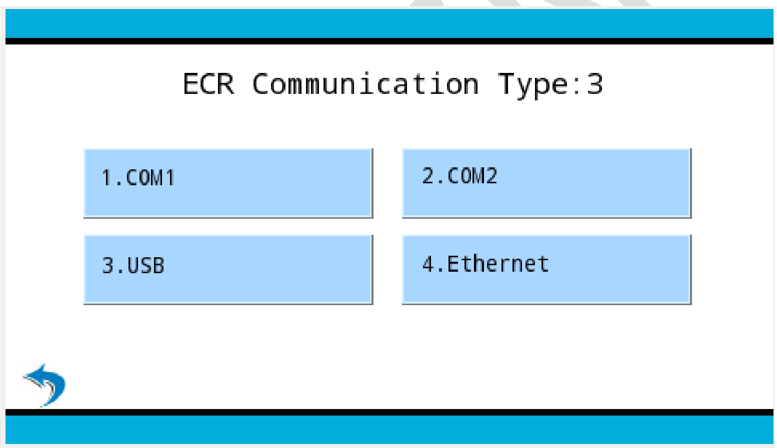
4. Enter same password as above.

5.5.2 UART/COM

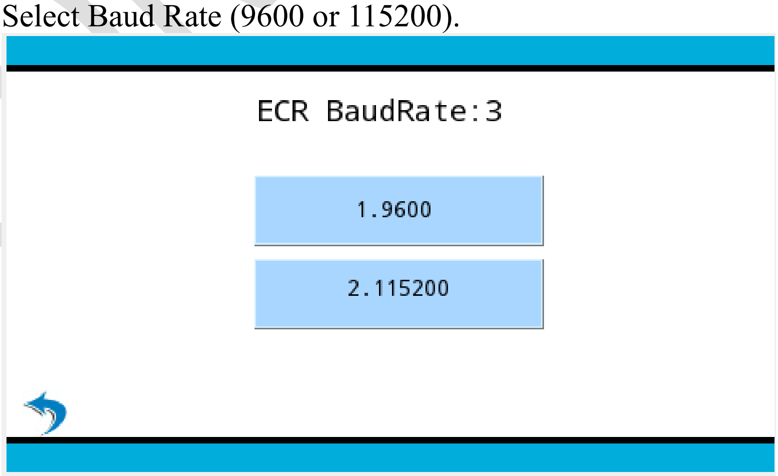
1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.



4. Select COM1 or COM2.

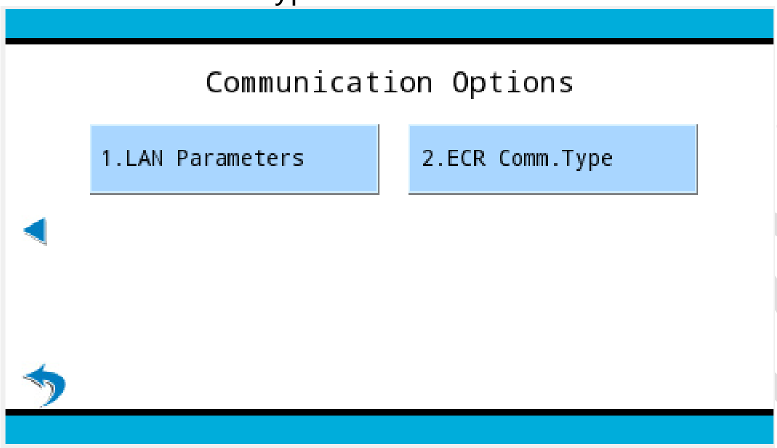


5. Select Baud Rate (9600 or 115200).

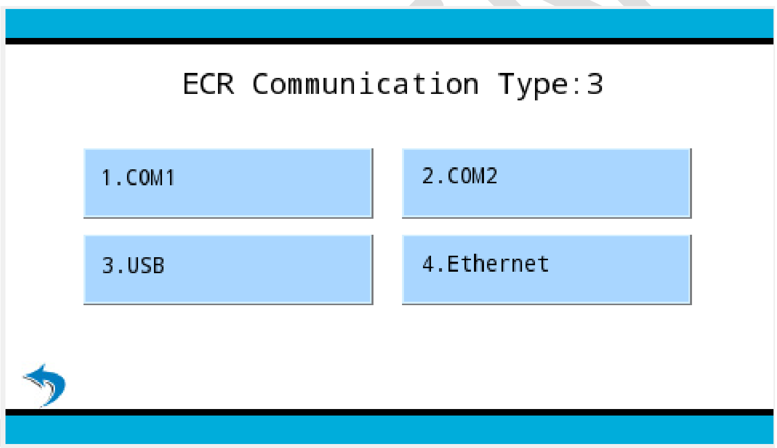


5.5.3 Ethernet – TCP/IP, HTTP, HTTPS, SSL

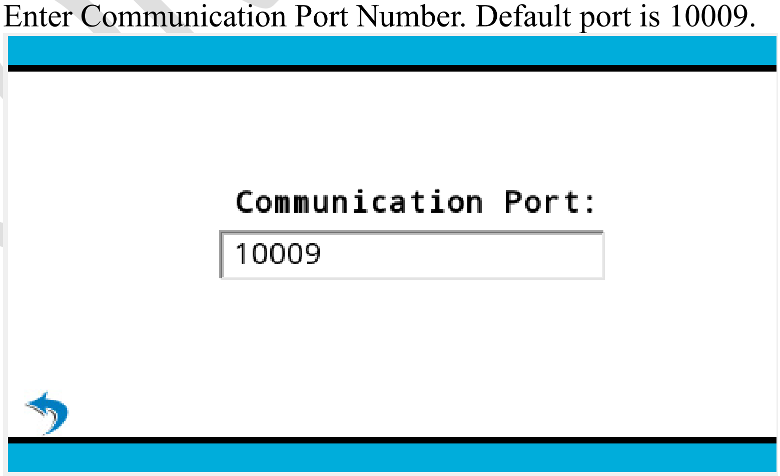
1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.



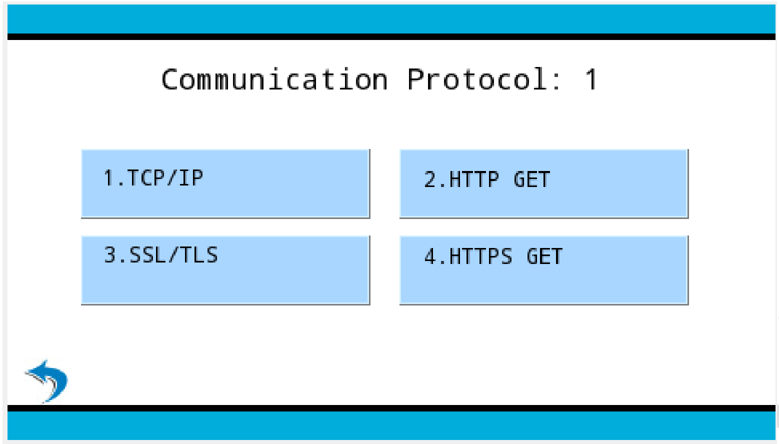
4. Select Ethernet.



5. Enter Communication Port Number. Default port is 10009.

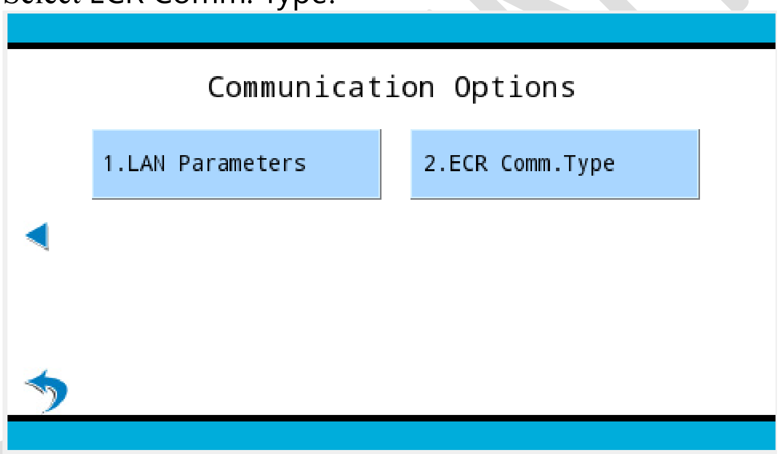


6. Choose Communication Protocol.

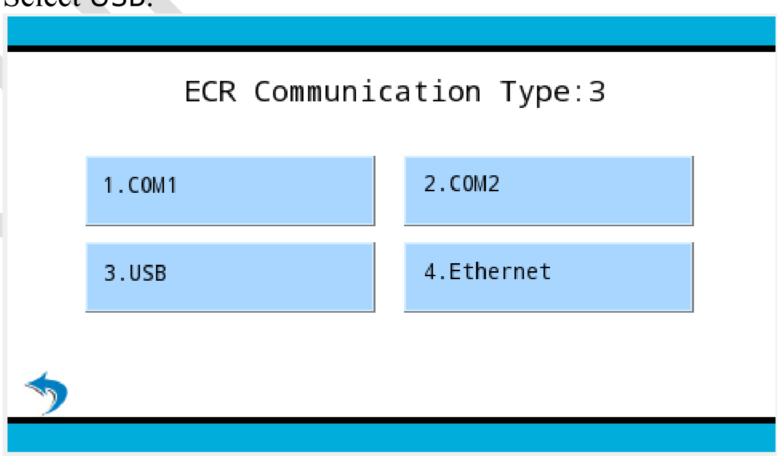


5.5.4 USB

1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.

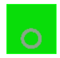


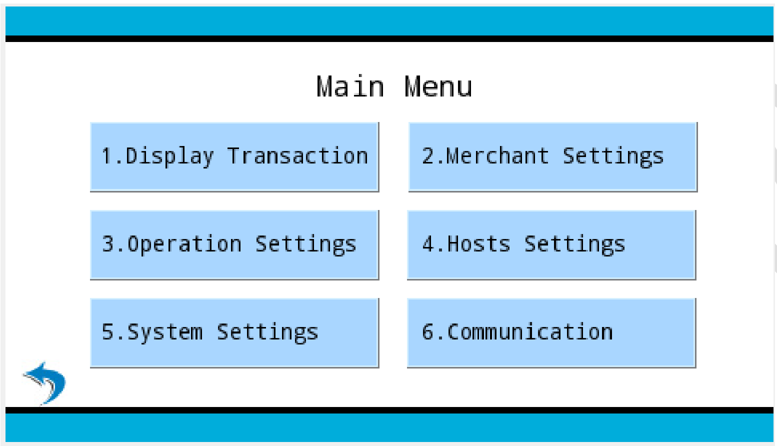
4. Select USB.



5.6 Px5/Px7

5.6.1 Access Communication Menu

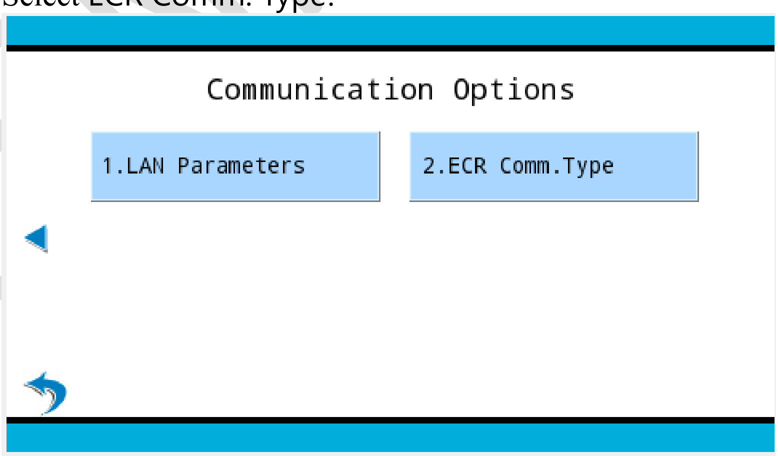
1. Press the Enter Key  and Num 1 Key 1 QZ . at the same time to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Select Communication.



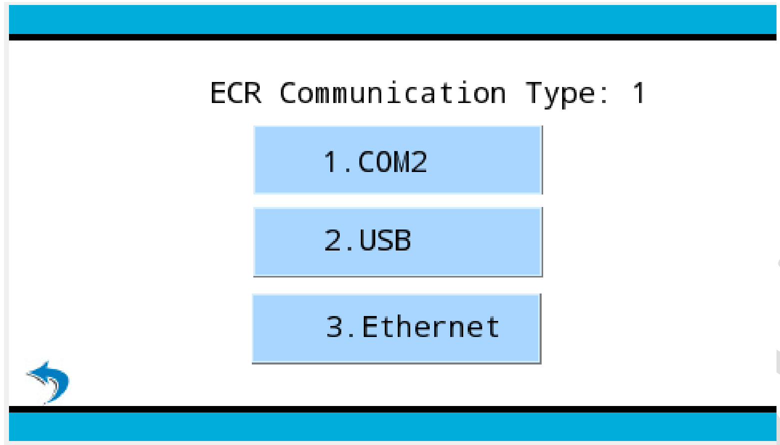
4. Enter same password as above.

5.6.2 UART/COM

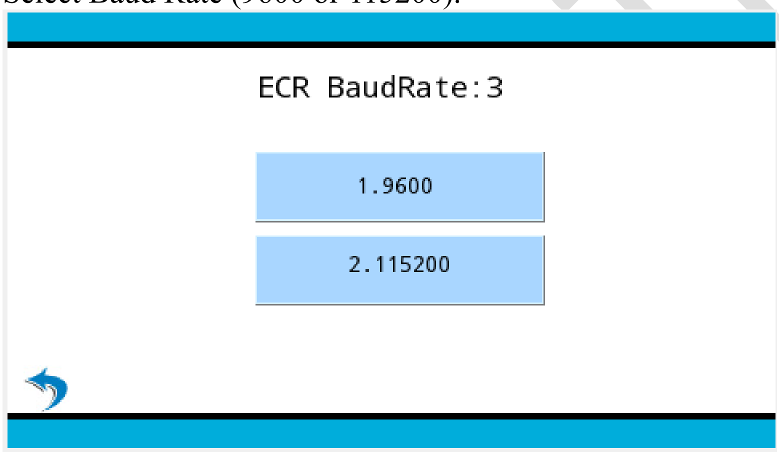
1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.



4. Select COM2.

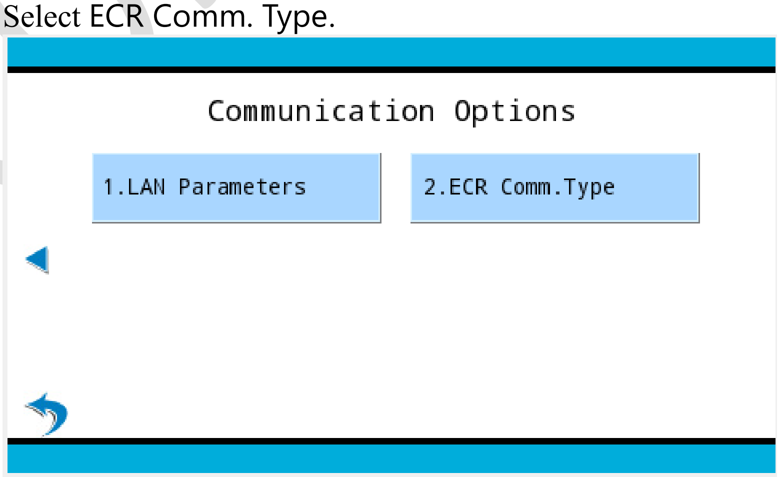


5. Select Baud Rate (9600 or 115200).

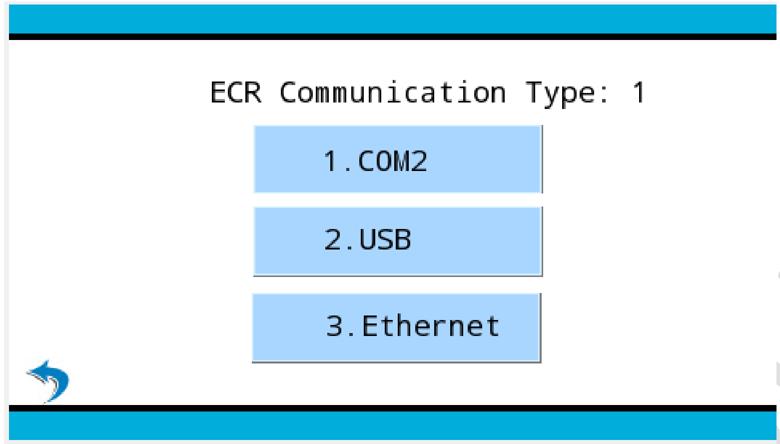


5.6.3 Ethernet – TCP/IP, HTTP, HTTPS, SSL

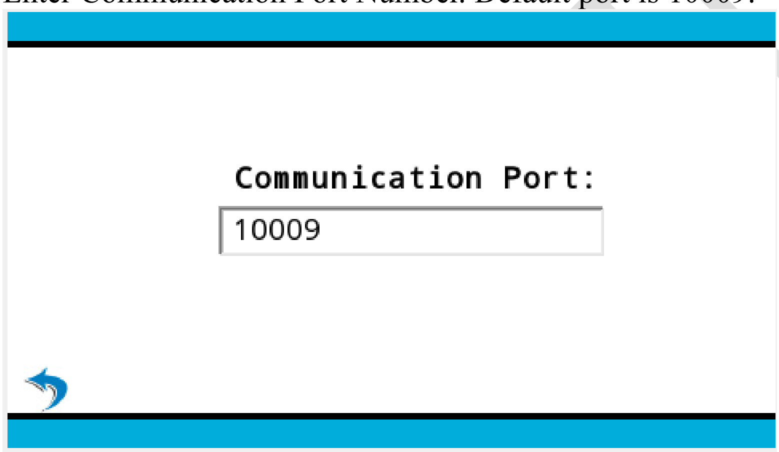
1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.



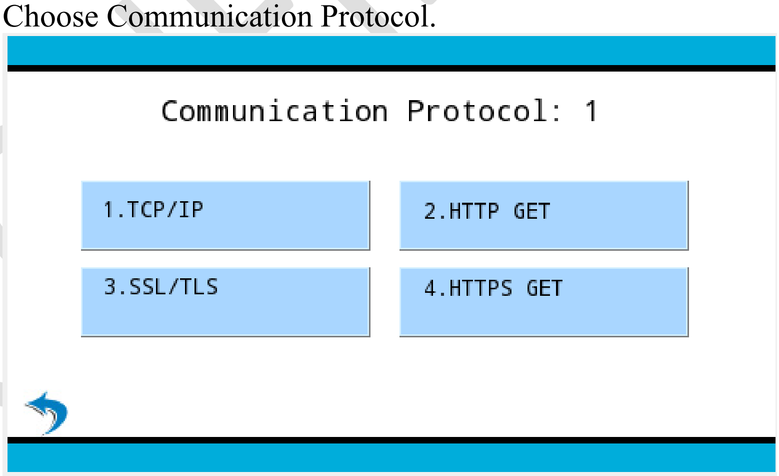
4. Select Ethernet.



5. Enter Communication Port Number. Default port is 10009.

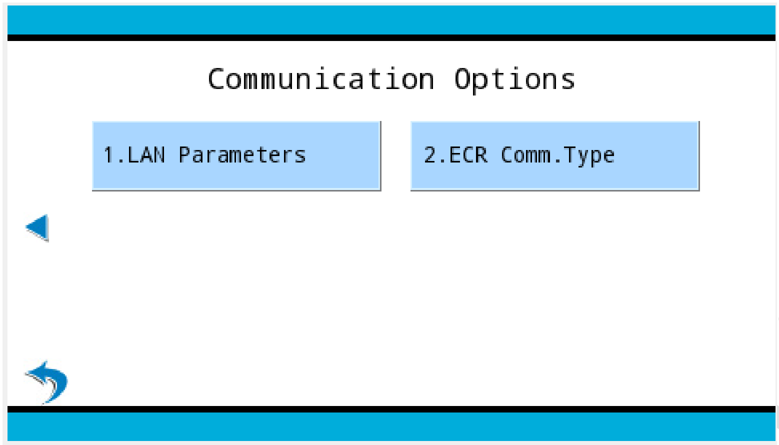


6. Choose Communication Protocol.

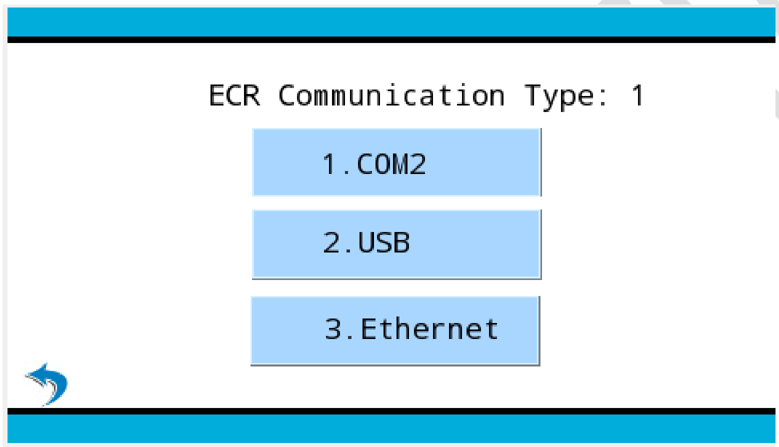


5.6.4 USB

1. Access Communication Menu
2. Scroll to the next page using the onscreen left arrow
3. Select ECR Comm. Type.


Communication Options
1.LAN Parameters
2.ECR Comm.Type

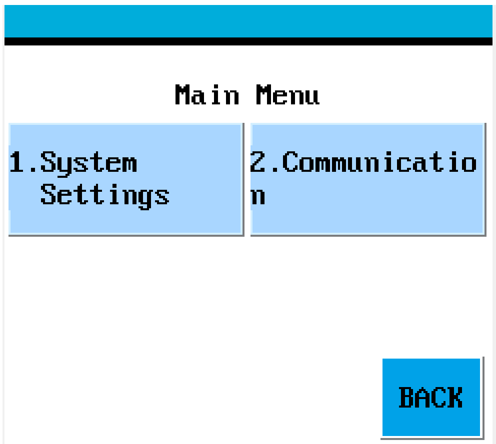
4. Select USB.


ECR Communication Type: 1
1.COM2
2.USB
3.Ethernet

5.7 S300

5.7.1 Access Communication Menu

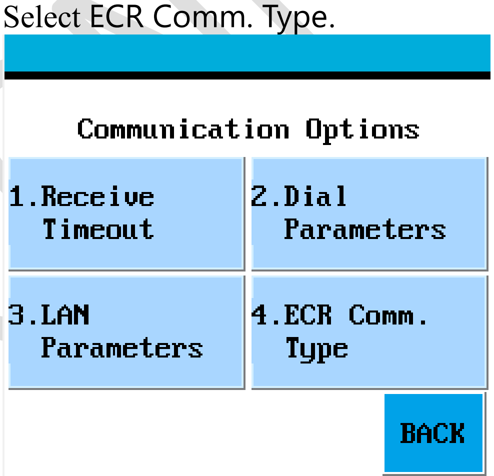
1. Press the Func Key and Num 1 Key at the same time to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Scroll down using the onscreen down arrow
4. Select Communication.



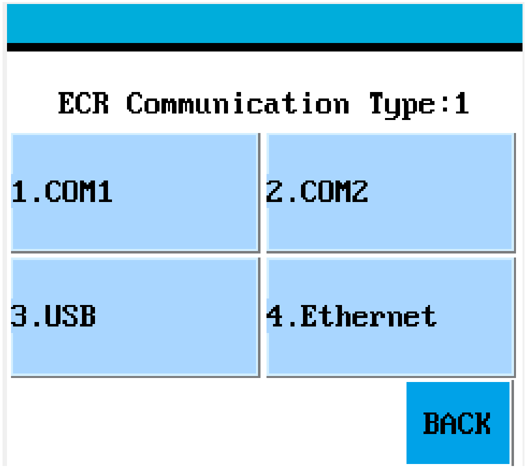
5. Enter same password as above.

5.7.2 UART/COM

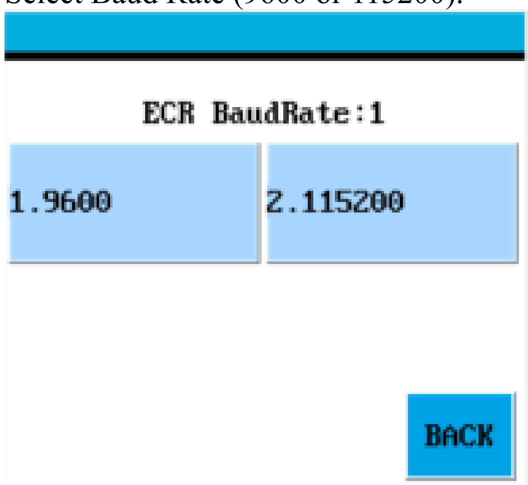
1. Access Communication Menu
2. Scroll down twice using the onscreen Down Arrow .
3. Select ECR Comm. Type.



4. Select COM1 or COM2.

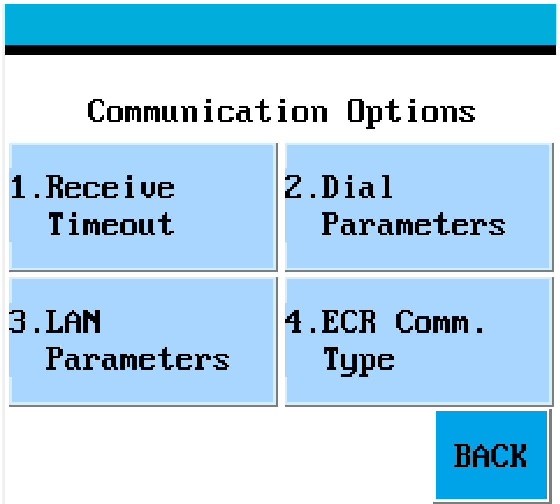


5. Select Baud Rate (9600 or 115200).

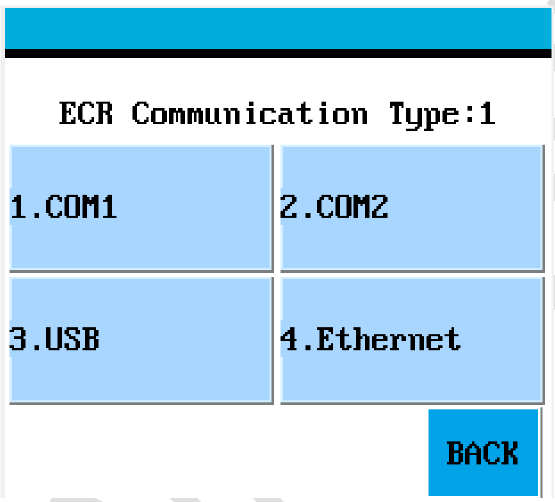


5.7.3 Ethernet – TCP/IP, HTTP, HTTPS, SSL

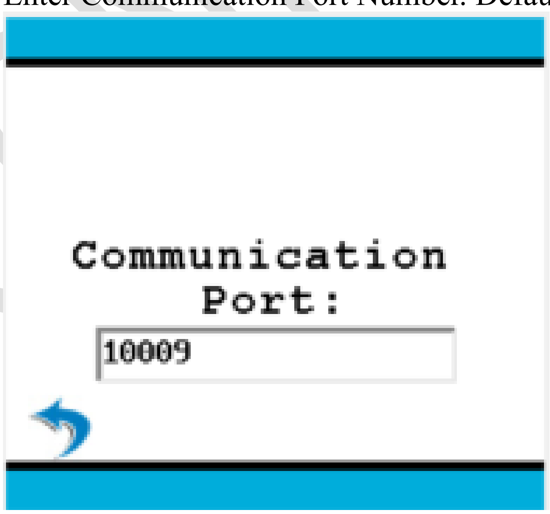
1. Access Communication Menu
2. Scroll down twice using the onscreen Down Arrow .
3. Select ECR Comm. Type.



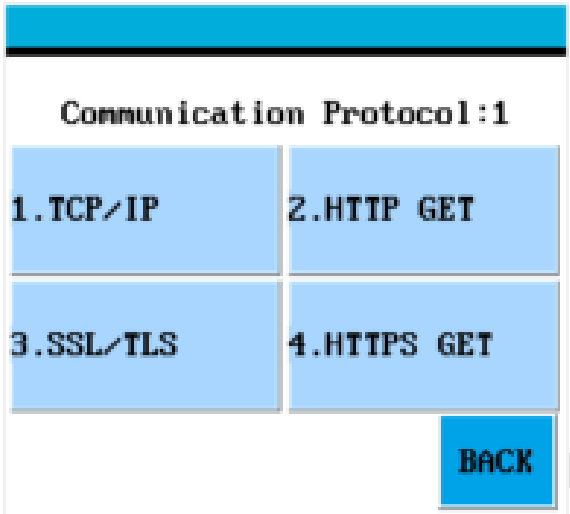
4. Select Ethernet.



5. Enter Communication Port Number. Default port is 10009.

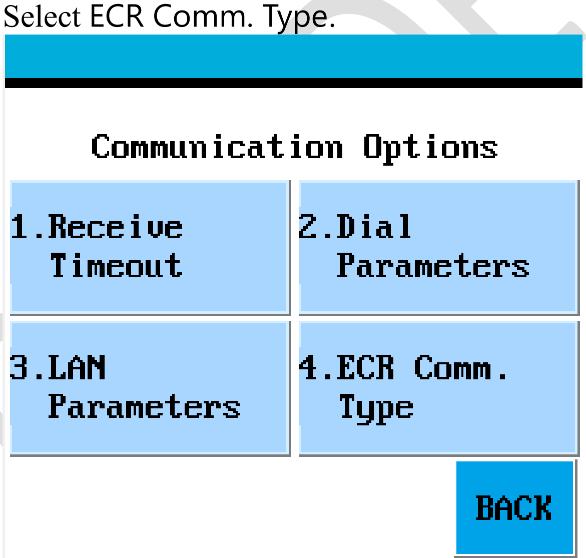


6. Choose Communication Protocol.

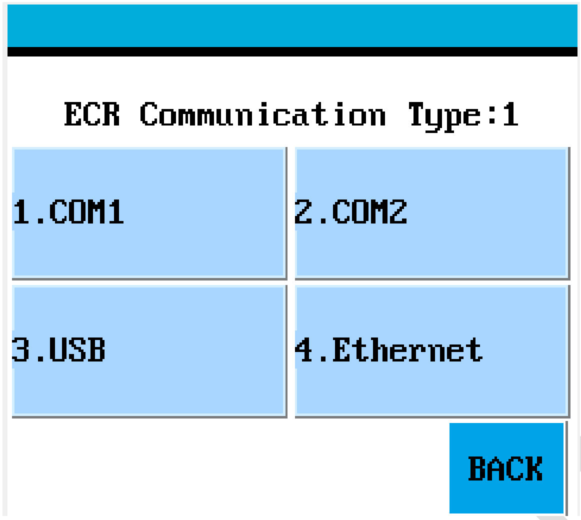


5.7.4 USB

1. Access Communication Menu
2. Scroll down twice using the onscreen Down Arrow .
3. Select ECR Comm. Type.



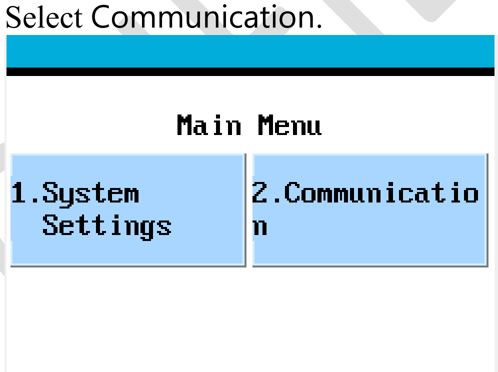
4. Select USB.



5.8 S920

5.8.1 Access Communication Menu

1. Press the Enter Key and Num 1 Key at the same time to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Scroll down using the onscreen down arrow
4. Select Communication.

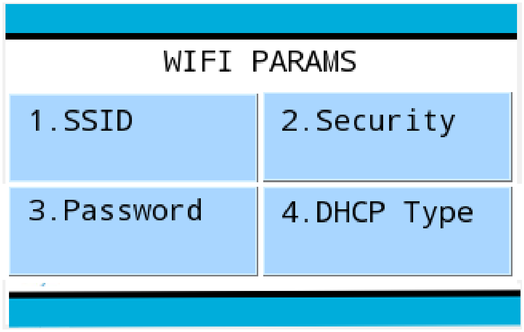


5. Enter same password as above.

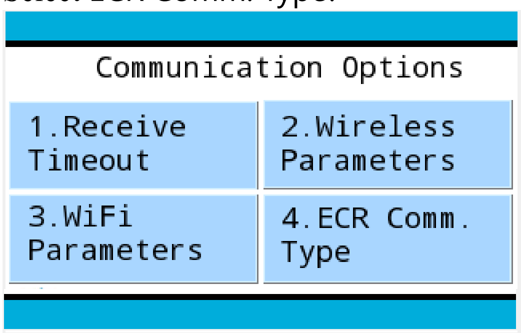
5.8.2 Ethernet – TCP/IP, HTTP, HTTPS, SSL

13. Access Communication Menu
14. Scroll down using the onscreen down arrow .
15. Select WiFi Parameters
16. Select SSID. Or if previously configured, scroll down using the onscreen down

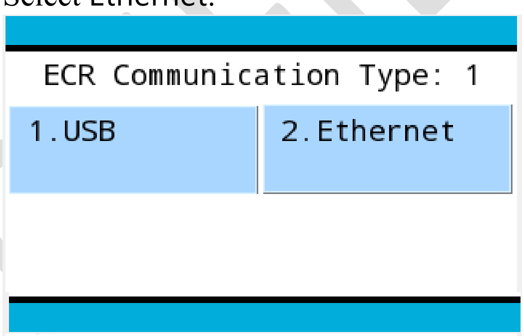
arrow and select Switch Router



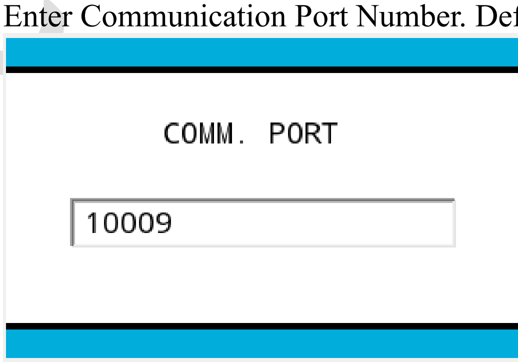
17. Choose the SSID of the WIFI router.
18. Select Password and enter Access Point Preshared Key
19. Press Cancel Key CANCEL
20. Scroll down twice using the Down Arrow Key .
21. Select ECR Comm. Type.



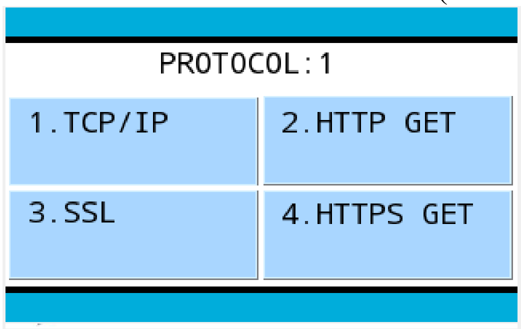
22. Select Ethernet.



23. Enter Communication Port Number. Default port is 10009.

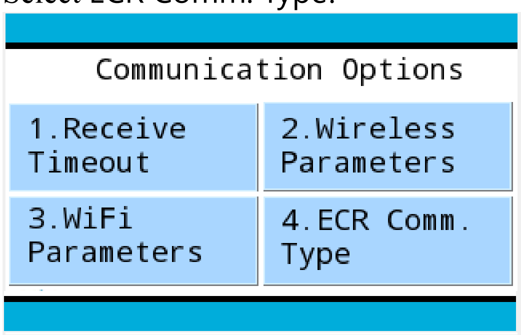


24. Choose Communication Protocol (TCP/IP, HTTP, HTTPS, SSL)

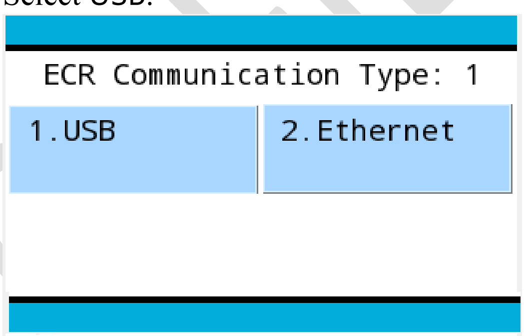


5.8.3 USB

1. Access Communication Menu
2. Scroll down twice using the On onscreen down arrow .
3. Select ECR Comm. Type.



4. Select USB.



5.9 SP30s

5.9.1 Access Communication Menu

1. Press Func Key FUNC to access Main Menu.
2. Enter Password. Default Password is 1 or today's date (format: mmddyyyy).
3. Scroll down using Down Key .
4. Select Communication.

MAIN MENU:

1.Display Trans.
2.Merchant Settings
3.Operation Settings
4.Host Settings
5.System Settings
6.Communication

5. Enter same password as above.

5.9.2 UART/COM

1. Access Communication Menu.
2. Scroll down using F2 Key F2.
3. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Ping Timeout
7.Dial Parameters
8.LAN Parameters
9.ECR Comm. Type

4. Select COM1.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.COM2

5. Select Baud Rate (9600 or 115200).

ECR BaudRate:1

1.9600
2.115200

5.9.3 Ethernet – TCP/IP, HTTP, HTTPS, SSL

1. Access Communication Menu.
2. Scroll down using Down Key .
3. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Ping Timeout
7.Dial Parameters
8.LAN Parameters
9.ECR Comm. Type

4. Select Ethernet.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.COM2

5. Enter Communication Port Number. Default port is 10009.

COMM. PORT:
10009

6. Choose Communication Protocol (TCP/IP, HTTP, HTTPS, SSL)

PROTOCOL: 1

1.TCP/IP
2.HTTP GET
3.SSL
4.HTTPS GET

5.9.4 USB

1. Access Communication Menu
2. Scroll down twice using the Down Key .
3. Select ECR Comm. Type.

COMM. OPTIONS

4.Connect Timeout
5.Receive Timeout
6.Ping Timeout
7.Dial Parameters
8.LAN Parameters
9.ECR Comm. Type

4. Select USB.

ECR COMM. TYPE:3

1.COM1
2.USB
3.Ethernet
4.COM2

6. HTTPS/SSL Communication Setup Requirement

6.1 PEM Files and Requirements

PAX Terminals support SSL communication for the POS system. The terminal needs to have the following SSL PEM certificate files loaded:

1. **serverkey.pem -** private key.

Supported Formats:

a. **PKCS#1 format:**
enclosed between "-----BEGIN RSA PRIVATE KEY-----" and "-----END RSA PRIVATE KEY-----" **[[Recommended]]**

b. **PKCS#8 format:**
enclosed between "-----BEGIN PRIVATE KEY-----" and "-----END PRIVATE KEY-----"
Limitation: browser must later than 20171103 ssl/https server will not be started successful lly for other formats.

c. **Converting Between Formats:**

   i. **Convert PKCS#1 to PKCS#8**
   $ openssl pkcs8 -topk8 -inform PEM -in pkcs1.pem -outform pem -nocrypt -out pkcs8.pem

   ii. **Convert PKCS#8 to PKCS#1**
   $ openssl rsa -in pkcs8.pem -out pkcs1.pem

2. **serverca.pem, servercert.pem**

serverca.pem – used by terminal to verify certificate from client, usually with empty text since current SSL server is built on PAX device is not verifying certificate from client.

servercert.pem – append all certificates defined in customer provided ca.pem to this file using format below. The first certificate should be the local certificate used by the terminal and the others are CA certificates.

Supported Formats:

a. enclosed between "-----BEGIN CERTIFICATE-----" and "-----END CERTIFICATE-----"
b. enclosed between "-----BEGIN X509 CERTIFICATE-----" and "-----END X509 CERTIFICATE-----"
c. enclosed between "-----BEGIN TRUSTED CERTIFICATE-----" and "-----END TRUSTED CERTIFICATE-----"

Note: ssl/https server will not be started successfully for other formats.

3. **Verifying that the private key matches a certificate**

$ openssl x509 -noout -modulus -in servercert.pem | openssl md5
$ openssl rsa -noout -modulus -in serverkey.pem | openssl md5

Note: MD5 value should be same, or otherwise the serverkey.pem will not match servercert.pem

4. Read certificate information

$ openssl x509 -in servercert.pem -noout -text

a. Issuer: who sign this certification, usually is CA
b. Subject : who own this certification
c. CA: True: this is a CA; False: this is not a CA

Below is result for command [openssl x509 -in servercert.pem -noout -text]

```
=======================================================================
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            48:e0:be:c3:86:9a:09:53:1a:d3:de:3d:7d:28:24:5c
    Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=GB, ST=Greater Manchester, L=Salford, O=COMODO CA Limited, CN=COMODO RSA Domain Validation Secure Server CA
        Validity
            Not Before: Jul 24 00:00:00 2015 GMT
            Not After : Jul 23 23:59:59 2018 GMT
        Subject: OU=Domain Control Validated, OU=PositiveSSL Wildcard, CN=*.econduitapps.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:dd:24:76:09:67:b4:3c:83:ad:1e:52:1b:aa:50:
                    af:96:79:80:cb:d8:97:53:76:bc:b2:4a:ca:e4:3d:
                    54:de:0e:5e:94:2b:6f:00:6e:4c:0e:94:ab:06:8e:
                    48:bb:39:33:69:2d:fe:f6:73:4b:6a:e7:c3:ed:de:
                    28:fc:e1:5a:14:f3:4f:8b:60:6d:79:7f:bc:0c:07:
                    e5:7c:54:52:6f:28:6c:85:9d:fc:bd:69:30:37:78:
                    84:79:88:0d:ec:94:9b:19:22:e7:bb:72:c7:ea:ee:
                    e8:fd:2c:6d:23:76:b7:52:31:75:74:ed:f7:4b:41:
                    8f:19:14:14:62:fc:65:a8:7a:87:e1:99:a1:d5:69:
                    3d:07:83:6a:dd:e6:97:79:d2:c1:30:80:d1:fb:05:
                    d6:b0:be:ad:63:fd:0f:77:7c:4d:35:b9:23:83:b4:
                    c4:d8:32:59:04:d3:72:42:49:6a:5f:8c:d5:b9:17:
                    61:6c:24:f2:b4:56:4f:59:89:c2:df:eb:ec:55:d1:
                    b6:7c:40:f5:a8:53:8e:e6:73:d4:23:2e:d7:05:7b:
                    de:b6:74:bb:7b:55:c2:d9:65:e7:68:55:06:67:96:
                    d3:87:41:26:ce:94:bb:17:e5:67:f6:3a:83:77:8c:
                    26:4d:ba:b6:26:90:e1:f6:35:22:2a:1d:49:34:b7:
                    74:5d
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                keyid:90:AF:6A:3A:94:5A:0B:D8:90:EA:12:56:73:DF:43:B4:3A:28:DA:E7

            X509v3 Subject Key Identifier:
                B7:49:7C:01:3B:06:68:56:48:BE:DC:DF:B1:AD:FD:D0:4B:6E:08:44
            X509v3 Key Usage: critical
                Digital Signature, Key Encipherment
```

```
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
            X509v3 Certificate Policies:
                Policy: 1.3.6.1.4.1.6449.1.2.2.7
                  CPS: https://secure.comodo.com/CPS
                Policy: 2.23.140.1.2.1

            X509v3 CRL Distribution Points:

                Full Name:
                  URI:http://crl.comodoca.com/COMODORSADomainValidationSecureServerCA.crl

            Authority Information Access:
                CA Issuers -
URI:http://crt.comodoca.com/COMODORSADomainValidationSecureServerCA.crt
                OCSP - URI:http://ocsp.comodoca.com

            X509v3 Subject Alternative Name:
                DNS:*.econduitapps.com, DNS:econduitapps.com
    Signature Algorithm: sha256WithRSAEncryption
         00:dc:62:da:42:52:15:8a:37:c3:21:3d:12:f8:58:bc:b3:0c:
         9e:01:8a:cb:ff:6c:33:f8:51:cf:6d:3c:41:c6:a1:7d:ca:91:
         7b:a3:6e:f5:d4:d1:8f:41:64:d8:1b:63:78:ac:a0:11:c0:69:
         65:3f:2a:09:60:20:40:fe:3b:68:77:53:af:ca:e7:fb:ac:35:
         c3:31:73:fe:de:55:31:82:94:ed:a3:2f:f8:64:91:e7:fc:83:
         91:63:08:47:0f:26:22:b3:d4:f5:fc:5a:48:c0:c5:5f:7c:8e:
         b1:c2:52:40:57:e2:4d:ec:0f:35:6a:13:56:6b:1e:2b:a1:f2:
         9b:b5:4b:ee:0d:8d:77:0d:f3:ce:1b:62:bc:4f:cc:56:c9:f4:
         dc:81:0c:ff:16:75:08:f8:23:87:ab:df:7a:11:97:99:4f:90:
         2c:8a:69:62:0e:10:46:51:b6:04:b1:6e:99:a5:69:6b:f0:e3:
         4a:d1:9f:3d:fe:15:af:13:25:b5:01:08:c1:b6:59:b4:ad:86:
         d2:cf:1b:3b:c2:e6:e1:56:0f:1c:74:b7:7b:ae:f8:ea:4d:08:
         1d:3a:06:ae:b7:b2:07:7a:5a:79:78:1f:cb:40:0c:ed:8f:a0:
         92:41:14:6c:9c:5f:f2:3c:1c:95:07:f9:a8:06:a1:0a:b2:78:
         4e:25:90:73
=================================================================================
```

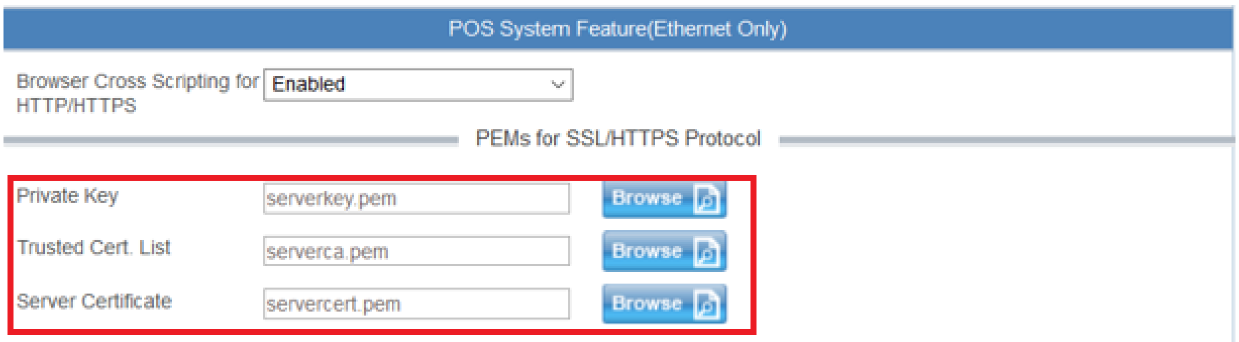
NOTE:

1. Key with password is not supported, you should either get a key without password from CA or convert the key with password to a key without password by using openssl rsa command. Exampe:
   $openssl rsa -in rsa_aes_private.key -passin pass:111111 -out rsa_private.key

6.2 Load PEM Files to PAX Terminal

The certificate files can be uploaded to the terminal via BroadPOS TMS.

1. In the Parameter Configuration Page, select Communication Tab
2. Under POS System Feature(Ethernet Only), there will be upload fields.
3. Browse and upload the corresponding certificate files to respective fields.



6.3 Generating a self-signed certificate

1. **Make demoCA directory**
   $ mkdir -p ./demoCA/{private,newcerts}
   $ touch ./demoCA/index.txt
   $ echo 01 > ./demoCA/serial
2. **Generate ca private key, RSA 2048 bit**
   $ openssl genrsa -out ./demoCA/private/cakey.pem 2048
3. **Create a self-signed ca**
   $ openssl req -new -x509 -extensions v3_ca-sha256 -days 3650 -key ./demoCA/private/cakey.pem
4. **Generate server private key, RSA 2048 bit**
   $ openssl genrsa -out serverkey.pem 2048
5. **Use server private key to create a Certificate Signing Request (CSR) file**
   $openssl req -new -sha256 -days 3650 -key serverkey.pem -out serverReq.pem
6. **Sign the server's CSR with demoCA, make sure [countryName, stateOrProvinceName, organizationName] match during CA creation**
   $ openssl ca -in serverReq.pem -out servercert.pem
7. **Copy serverca.pem to current directory**
   $ cp ./demoCA/cacert.pem serverca.pem

6.4 Import Self-Signed CA to Trusted Root Certification Authorities (Windows Only):

1. Win -> mmc.exe
2. Select File, and click **Add/Remove Snap-in...**
3. Select the **Certificates** snap-in, and click **Add**
4. Select **Computer account**, and click **Next**
5. Select **Local computer**, and click **Finish**
6. Click **OK**
7. In the left pane, expand **Certificates (Local Computer)**
8. Expand **Trusted Root Certification Authorities**, and click **Certificate**
9. Right-click on the newly created certificate, select **All Tasks**, and click **Import…**
10. The Certificate Export Wizard will open. Click **Next** to continue.
11. Specify a file name, and click **Next**.
12. Click **Next**
13. Click **Finish**